8.5 For the current-steering circuit of Fig. P8.5, find I_O in terms of I_{REF} and device W/L ratios.

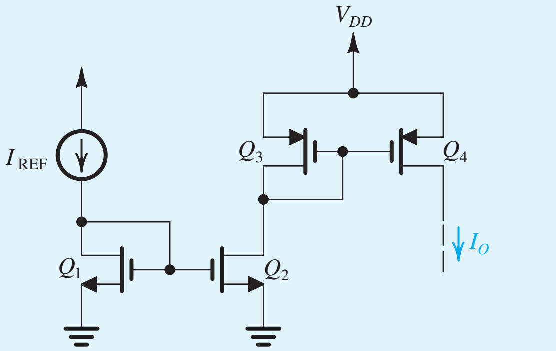


Figure P8.5

D 8.6 The current-steering circuit of Fig. P8.6 is fabricated in a CMOS technology for which $\mu_n C_{ox} = 400\ \mu\text{A/V}^2$, $\mu_p C_{ox} = 100\ \mu\text{A/V}^2$, $V_{tn} = 0.5$ V $V_{tp} = -0.5$ V, $V'_{An} = 5$ V/μ m, and $|V'_{Ap}| = 5$ V/μm. If all devices have $L = 0.5\ \mu$m, design the circuit so that $I_{\text{REF}} = 20\ \mu$A, $I_2 = 100\ \mu$A, $I_3 = I_4 = 40\ \mu$A, and $I_5 = 80\ \mu$A. Use the minimum possible device widths needed to achieve proper operation of the current source Q_2 for voltages at its drain as high as $+0.8$ V and proper operation of the current sink Q_5 with voltages at its drain as low as -0.8 V. Specify the widths of all devices and the value of R. Find the output resistance of the current source Q_2 and the output resistance of the current sink Q_5.

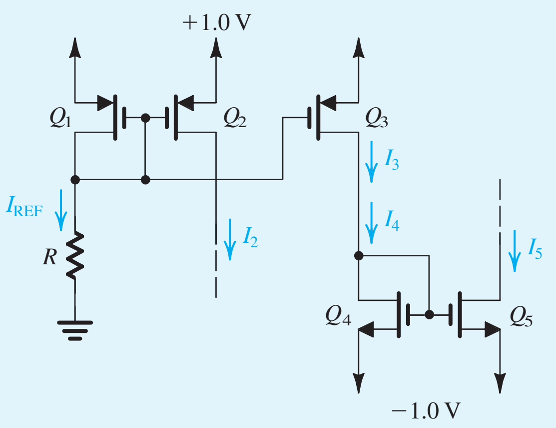


Figure P8.6

***8.7** A PMOS current mirror consists of three PMOS transistors, one diode connected and two used as current outputs. All transistors have $|V_t| = 0.6$ V, $k'_p = 100\ \ \mu\text{A/V}^2$, and $L = 1.0\ \mu$m but three different widths, namely, 10 μm, 20 μm, and 40 μm. When the diode-connected transistor is supplied from a 100-μA source, how many different output currents are available? Repeat with two of the transistors diode connected and the third used to provide current output. For each possible input-diode combination, give the values of the output currents and of the V_{SG} that results.

8.8 Consider the basic bipolar current mirror of Fig. 8.7 for the case in which Q_1 and Q_2 are identical devices having $I_S = 10^{-17}$A.

(a) Assuming the transistor β is very high, find the range of V_{BE} and I_O corresponding to I_{REF} increasing from 10 μA to 10 mA. Assume that Q_2 remains in the active mode, and neglect the Early effect.

(b) Find the range of I_O corresponding to I_{REF} in the range of 10 μA to 10 mA, taking into account the finite β. Assume that β remains constant at 100 over the current range 0.1 mA to 5 mA but that at $I_C \simeq 10$ mA and at $I_C \simeq 10\ \mu$A, $\beta = 50$. Specify I_O corresponding to $I_{\text{REF}} = 10\ \mu$A, 0.1 mA, 1 mA, and 10 mA. Note that β variation with current causes the current transfer ratio to vary with current.

8.9 Consider the basic BJT current mirror of Fig. 8.7 for the case in which Q_2 has m times the area of Q_1. Show that the current transfer ratio is given by Eq. (8.19). If β is specified to be a minimum of 80, what is the largest current transfer ratio possible if the error introduced by the finite β is limited to 10%?

8.10 Give the circuit for the *pnp* version of the basic current mirror of Fig. 8.7. If β of the *pnp* transistor is 50, what is the current gain (or transfer ratio) I_O/I_{REF} for the case of identical transistors, neglecting the Early effect?

8.11 Consider the basic BJT current mirror of Fig. 8.7 when Q_1 and Q_2 are matched and $I_{\text{REF}} = 1$ mA. Neglecting the effect of finite β, find the change in I_O, both as an absolute value and as a percentage, corresponding to V_O changing from 1 V to 10 V. The Early voltage is 90 V.

D 8.12 The current-source circuit of Fig. P8.12 utilizes a pair of matched *pnp* transistors having $I_S = 10^{-15}$A, $\beta = 50$, and $|V_A| = 50$ V. It is required to design the circuit to provide an output current $I_O = 1$ mA at $V_O = 1$ V. What values of I_{REF} and R are needed? What is the maximum allowed value of V_O while the current source continues to operate properly? What change occurs in I_O corresponding to V_O changing from the

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

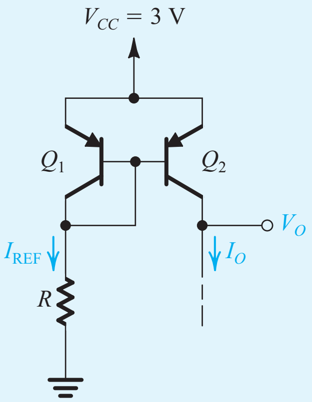


Figure P8.12

maximum positive value to –5 V? *Hint:* Adapt Eq. (8.21) for this case as:

$$I_O = I_{REF}\left[\frac{1+\dfrac{3-V_O-V_{EB}}{|V_A|}}{1+\dfrac{2}{\beta}}\right]$$

8.13 Find the voltages at all nodes and the currents through all branches in the circuit of Fig. P8.13. Assume $|V_{BE}| = 0.7$ V and $\beta = \infty$.

8.14 For the circuit in Fig. P8.14, let $|V_{BE}| = 0.7$ V and $\beta = \infty$. Find I, V_1, V_2, V_3, V_4, and V_5 for (a) $R = 10$ kΩ and (b) $R = 100$ kΩ.

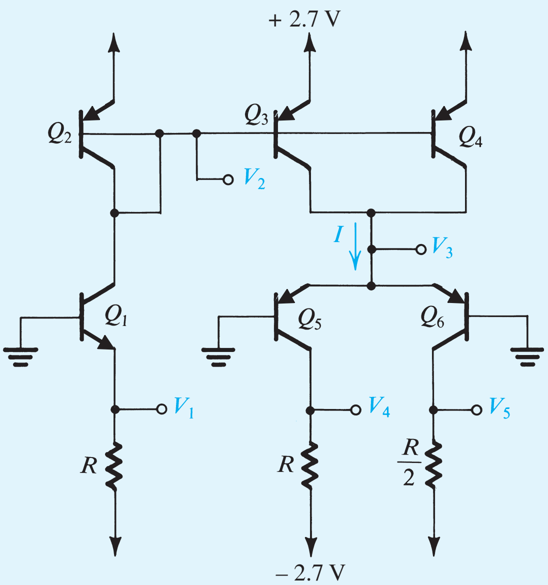


Figure P8.14

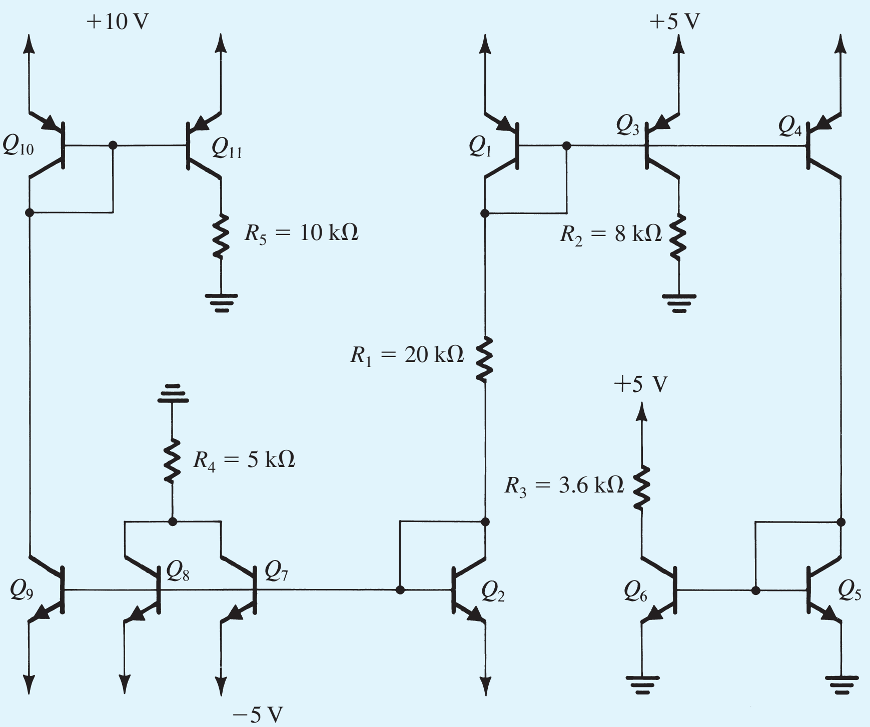


Figure P8.13

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

D 8.15 Using the ideas embodied in Fig. 8.10, design a multiple-mirror circuit using power supplies of ± 5 V to create source currents of 0.2 mA, 0.4 mA, and 0.8 mA and sink currents of 0.5 mA, 1 mA, and 2 mA. Assume that the BJTs have $|V_{BE}| \simeq 0.7$ V and large β. What is the total power dissipated in your circuit?

***8.16** The circuit shown in Fig. P8.16 is known as a **current conveyor**.

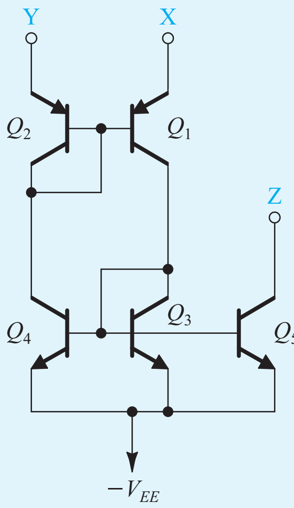


Figure P8.16

(a) Assuming that Y is connected to a voltage V, a current I is forced into X, and terminal Z is connected to a voltage that keeps Q_5 in the active region, show that a current equal to I flows through terminal Y, that a voltage equal to V appears at terminal X, and that a current equal to I flows through terminal Z. Assume β to be large; corresponding transistors are matched, and all transistors are operating in the active region.

(b) With Y connected to ground, show that a virtual ground appears at X. Now, if X is connected to a +5-V supply through a 10-kΩ resistor, what current flows through Z?

8.17 The MOSFETs in the current mirror of Fig. 8.12(a) have equal channel lengths of 0.5 μm, $W_1 = 10$ μm, $W_2 = 50$ μm, $\mu_n C_{ox} = 500$ μA/V^2, and $V_A' = 10$ V/μm. If the input bias current is 100 μA, find R_{in}, A_{is}, and R_o.

D 8.18 The MOSFETs in the current mirror of Fig. 8.12(a) have equal channel lengths, $\mu_n C_{ox} = 400$ μA/V^2 and $V_A' = 20$ V/μm. If the input bias current is 200 μA, find W_1, W_2, and L to obtain a short-circuit current gain of 4, an input resistance of 500 Ω, and an output resistance of 20 kΩ.

8.19 Figure P8.19 shows an amplifier utilizing a current mirror Q_2–Q_3. Here Q_1 is a common-source amplifier fed with $v_I = V_{GS} + v_i$, where V_{GS} is the gate-to-source dc bias voltage of Q_1 and v_i is a small signal to be amplified. Find the signal component of the output voltage v_O and hence the small-signal voltage gain v_o/v_i. Also, find the small-signal resistance of the diode-connected transistor Q_2 in terms of g_{m2}, and hence the total resistance between the drain of Q_1 and ground. What is the voltage gain of the CS amplifier Q_1? Neglect all r_o's.

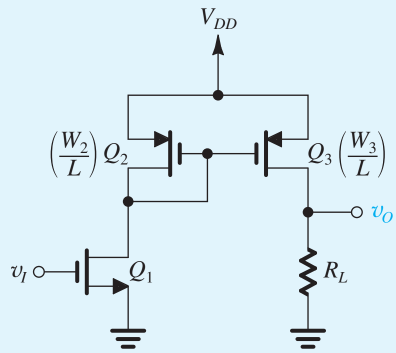


Figure P8.19

***8.20** Figure P8.20 shows a current-mirror circuit prepared for small-signal analysis. Replace the BJTs with their hybrid-π models and find expressions for R_{in}, i_o/i_i, and R_o, where i_o is the output short-circuit current. Assume $r_o \gg r_\pi$.

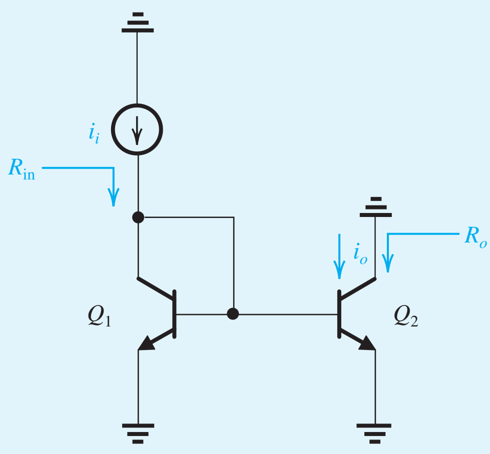


Figure P8.20

8.21 It is required to find the incremental (i.e., small-signal) resistance of each of the diode-connected transistors shown in

Fig. P8.21. Assume that the dc bias current $I = 0.1$ mA. For the MOSFET, let $\mu_n C_{ox} = 200$ μA/V^2 and $W/L = 10$. Neglect r_o for both devices.

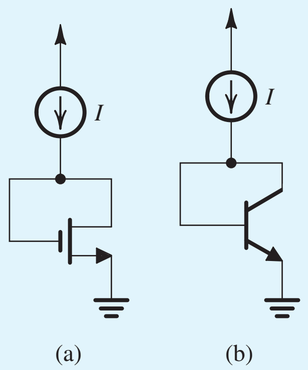


Figure P8.21

8.22 For the base-current-compensated mirror of Fig. 8.11, let the three transistors be matched and specified to have a collector current of 1 mA at $V_{BE} = 0.7$ V. For I_{REF} of 100 μA and assuming $\beta = 100$, what will the voltage at node x be? If I_{REF} is increased to 1 mA, what is the change in V_x? What is the value of I_O obtained with $V_O = V_x$ in both cases? Give the percentage difference between the actual and ideal value of I_O. What is the lowest voltage at the output for which proper current-source operation is maintained?

D 8.23 Extend the current-mirror circuit of Fig. 8.11 to n outputs. What is the resulting current transfer ratio from the input to each output, I_O/I_{REF}? If the deviation from unity is to be kept at 0.2% or less, what is the maximum possible number of outputs for BJTs with $\beta = 150$?

***8.24** For the base-current-compensated mirror of Fig. 8.11, show that the incremental input resistance (seen by the reference current source) is approximately $2V_T/I_{REF}$. Evaluate R_{in} for $I_{REF} = 100$ μA. (*Hint*: Q_3 is operating at a current $I_{E3} = 2I_C/\beta$, where I_C is the operating current of each of Q_1 and Q_2. Replace each transistor with its T model and neglect r_o.)

Section 8.3: The Basic Gain Cell

8.25 Find g_m, r_π, r_o, and A_0 for the CE amplifier of Fig. 8.13(b) when operated at $I = 10$ μA, 100 μA, and 1 mA. Assume $\beta = 100$ and remains constant as I is varied, and that $V_A = 10$ V. Present your results in a table.

8.26 Consider the CE amplifiers of Fig. 8.13(b) for the case of $I = 0.5$ mA, $\beta = 100$, and $V_A = 100$ V. Find R_{in}, A_{vo}, and R_o. If it is required to raise R_{in} by a factor of 5 by changing I, what value of I is required, assuming that β remains unchanged? What are the new values of A_{vo} and R_o? If the amplifier is fed with a signal source having $R_{sig} = 5$ kΩ and is connected to a load of 100-kΩ resistance, find the overall voltage gain, v_o/v_{sig}.

8.27 Find the intrinsic gain of an NMOS transistor fabricated in a process for which $k_n' = 400$ μA/V^2 and $V_A' = 10$ V μm. The transistor has a 0.5-μm channel length and is operated at $V_{OV} = 0.2$ V. If a 2-mA/V transconductance is required, what must I_D and W be?

8.28 An NMOS transistor fabricated in a certain process is found to have an intrinsic gain of 50 V/V when operated at an I_D of 100 μA. Find the intrinsic gain for $I_D = 25$ μA and $I_D = 400$ μA. For each of these currents, find the factor by which g_m changes from its value at $I_D = 100$ μA.

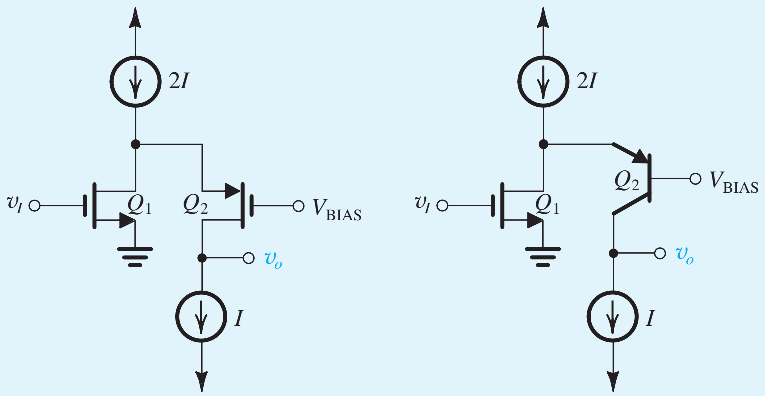
D 8.29 Consider an NMOS transistor fabricated in a 0.18-μm technology for which $k_n' = 400$ μA/V^2 and $V_A' = 5$ V/μm. It is required to obtain an intrinsic gain of 20 V/V and a g_m of 2 mA/V. Using $V_{OV} = 0.2$ V, find the required values of L, W/L, and the bias current I.

D 8.30 Sketch the circuit for a current-source-loaded CS amplifier that uses a PMOS transistor for the amplifying device. Assume the availability of a single +1.8-V dc supply. If the transistor is operated with $|V_{OV}| = 0.2$ V, what is the highest instantaneous voltage allowed at the drain?

8.31 An NMOS transistor operated with an overdrive voltage of 0.25 V is required to have a g_m equal to that of an *npn* transistor operated at $I_C = 0.1$ mA. What must I_D be? What value of g_m is realized?

8.32 For an NMOS transistor with $L = 1$ μm fabricated in the 0.5-μm process specified in Table J.1 in Appendix J, find g_m, r_o, and A_0 if the device is operated with $V_{OV} = 0.5$ V and $I_D = 100$ μA. Also, find the required device width W.

8.33 For an NMOS transistor with $L = 0.3$ μm fabricated in the 0.18-μm process specified in Table J.1 in Appendix J, find g_m, r_o, and A_0 obtained when the device is operated at $I_D = 100$ μA with $V_{OV} = 0.2$ V. Also, find W.



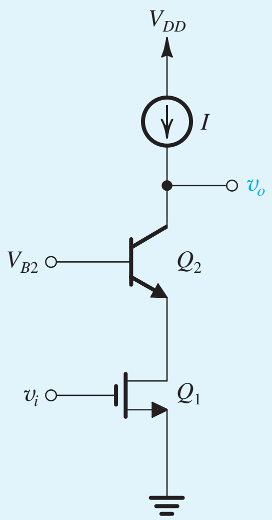
(c) (d)

Figure P8.80 *continued*

Let $I = 100\ \mu\text{A}$, and assume that the MOSFETs are operating at $|V_{OV}| = 0.2$ V. Assume the current sources are ideal. For each circuit determine R_{in}, R_o, and A_{vo}. Comment on your results.

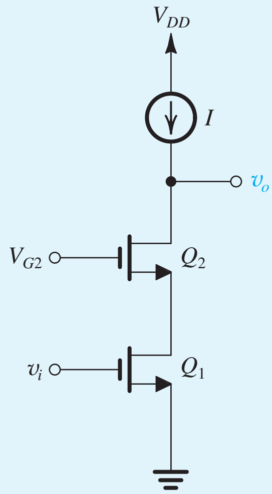
8.81 In this problem, we will explore the difference between using a BJT as cascode device and a MOSFET as cascode device. Refer to Fig. P8.81. Given the following data, calculate G_m, R_o, and A_{vo} for the circuits (a) and (b):

$I = 100\ \mu\text{A}$, $\beta = 125$, $\mu_n C_{ox} = 400\ \mu\text{A/V}^2$, $W/L = 25$, $V_A = 1.8$ V



(a)

Figure P8.81



(b)

Figure P8.81 *continued*

Section 8.6: Current-Mirror Circuits with Improved Performance

SIM **8.82** In a particular cascoded current mirror, such as that shown in Fig. 8.39, all transistors have $V_t = 0.6$ V, $\mu_n C_{ox} = 160\ \mu\text{A/V}^2$, $L = 1\ \mu\text{m}$, and $V_A = 10$ V. Width $W_1 = W_4 = 4\ \mu\text{m}$, and $W_2 = W_3 = 40\ \mu\text{m}$. The reference current I_{REF} is $20\ \mu\text{A}$. What output current results? What are the voltages at the gates of Q_2 and Q_3? What is the lowest voltage at the output for which current-source operation is possible? What are the values of g_m and

r_o of Q_2 and Q_3? What is the output resistance of the mirror?

8.83 Find the output resistance of the double-cascode current mirror of Fig. P8.83.

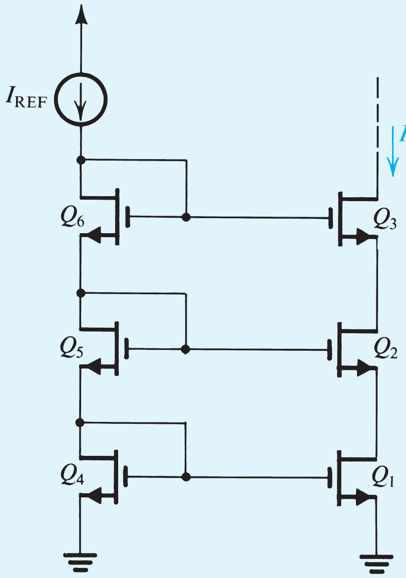


Figure P8.83

8.84 Consider the Wilson current-mirror circuit of Fig. 8.40 when supplied with a reference current I_{REF} of 1 mA. What is the change in I_O corresponding to a change of +10 V in the voltage at the collector of Q_3? Give both the absolute value and the percentage change. Let $\beta = 100$ and $V_A = 100$ V.

D 8.85 (a) The circuit in Fig. P8.85 is a modified version of the Wilson current mirror. Here the output transistor is "split" into two matched transistors, Q_3 and Q_4. Find I_{O1} and I_{O2} in terms of I_{REF}. Assume all transistors to be matched with current gain β.
(b) Use this idea to design a circuit that generates currents of 0.1 mA, 0.2 mA, and 0.4 mA, using a reference current source of 0.7 mA. What are the actual values of the currents generated for $\beta = 50$?

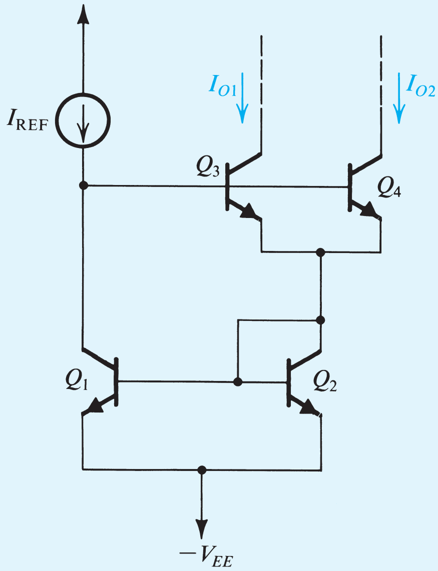


Figure P8.85

D 8.86 Use the *pnp* version of the Wilson current mirror to design a 0.1-mA current source. The current source is required to operate with the voltage at its output terminal as low as –2.5 V. If the power supplies available are ±2.5 V, what is the highest voltage possible at the output terminal?

***8.87** For the Wilson current mirror of Fig. 8.40, show that the incremental input resistance seen by I_{REF} is approximately $2V_T/I_{REF}$. (Neglect the Early effect in this derivation and assume a signal ground at the output.) Evaluate R_{in} for $I_{REF} =$ 0.2 mA.

***8.88** Consider the Wilson MOS mirror of Fig. 8.41(a) for the case of all transistors identical, with $W/L = 10$, $\mu_n C_{ox} =$ 400 μA/V^2, $V_{tn} = 0.5$ V, and $V_A = 18$ V. The mirror is fed with $I_{REF} = 180$ μA.

(a) Obtain an estimate of V_{OV} and V_{GS} at which the three transistors are operating, by neglecting the Early effect.
(b) Noting that Q_1 and Q_2 are operating at different V_{DS}, obtain an approximate value for the difference in their currents and hence determine I_O.
(c) To eliminate the systematic error between I_O and I_{REF} caused by the difference in V_{DS} between Q_1 and Q_2, a diode-connected transistor Q_4 can be added to the circuit

as shown in Fig. 8.41(c). What do you estimate I_O now to be?

(d) What is the minimum allowable voltage at the output node of the mirror?

(e) Convince yourself that Q_4 will have no effect on the output resistance of the mirror. Find R_o.

(f) What is the change in I_O (both absolute value and percentage) that results from $\Delta V_O = 1$ V?

8.89 Show that the incremental input resistance (seen by I_{REF}) for the Wilson MOS mirror of Fig. 8.41(a) is $2/g_m$. Assume that all three transistors are identical and neglect the Early effect. Also, assume a signal ground at the output. (*Hint*: Replace all transistors by their T model and remember that Q_1 is equivalent to a resistance $1/g_m$.)

D 8.90 (a) Utilizing a reference current of 200 μA, design a Widlar current source to provide an output current of 20 μA. Assume β to be high.

(b) If $\beta = 200$ and $V_A = 50$ V, find the value of the output resistance, and find the change in output current corresponding to a 5-V change in output voltage.

D 8.91 Design three Widlar current sources, each having a 100-μA reference current: one with a current transfer ratio of 0.8, one with a ratio of 0.10, and one with a ratio of 0.01, all assuming high β. For each, find the output resistance, and contrast it with r_o of the basic unity-ratio source that is providing the desired current and for which $R_E = 0$. Use $\beta = \infty$ and $V_A = 50$ V.

D 8.92 (a) For the circuit in Fig. P8.92, assume BJTs with high β and $v_{BE} = 0.7$ V at 1 mA. Find the value of R that will result in $I_O = 10$ μA.

(b) For the design in (a), find R_o assuming $\beta = 100$ and $V_A = 40$ V.

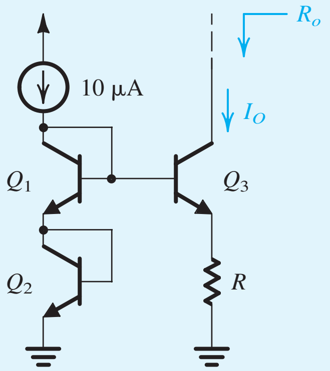


Figure P8.92

D 8.93 If the *pnp* transistor in the circuit of Fig. P8.93 is characterized by its exponential relationship with a scale current I_S, show that the dc current I is determined by $IR = V_T \ln(I/I_S)$. Assume Q_1 and Q_2 to be matched and Q_3, Q_4, and Q_5 to be matched. Find the value of R that yields a current $I = 200$ μA. For the BJT, $V_{EB} = 0.7$ V at $I_E = 1$ mA.

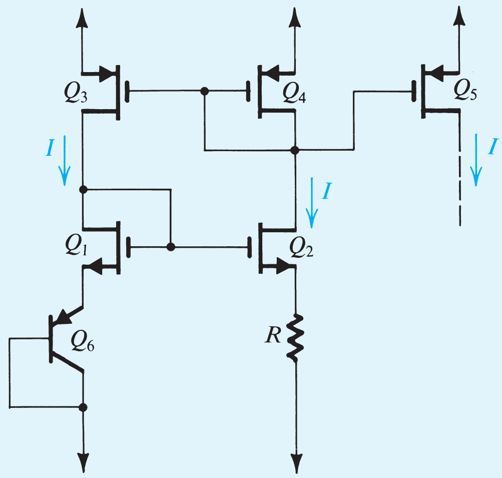


Figure P8.93

Section 8.7: Some Useful Transistor Pairings

8.94 Use the source-follower equivalent circuit in Fig. 8.45(b) to show that its output resistance is given by

$$R_o = r_{o3} \| r_{o1} \| \frac{1}{g_m + g_{mb}} \simeq \frac{1}{g_m + g_{mb}}$$

8.95 A source follower for which $k_n' = 200$ μA/V^2, $V_A' = 20$ V/μm, $\chi = 0.2$, $L = 0.5$ μm, $W = 20$ μm, and $V_t = 0.6$ V is required to provide a dc level shift (between input and output of 0.9 V.) What must the bias current be? Find g_m, g_{mb}, r_o, A_{vo}, and R_o. Assume that the bias current source has an output resistance equal to r_o. Also find the voltage gain when a load resistance of 2 kΩ is connected to the output.

8.96 The transistors in the circuit of Fig. P8.96 have $\beta = 100$ and $V_A = 50$ V.